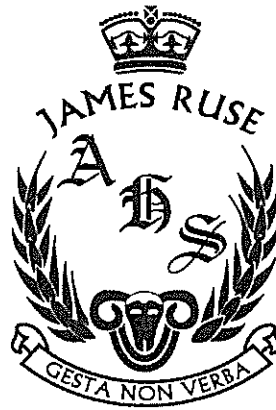


Name:	
Class:	



TRIAL HIGHER SCHOOL CERTIFICATE EXAMINATION 2012

MATHEMATICS EXTENSION 2

General Instructions:

- Reading Time: 5 minutes.
- Working Time: 3 hours.
- Write in black or blue pen.
- Board approved calculators & templates may be used
- A Standard integral Sheet is provided.
- In every question, show all necessary working
- Marks may not be awarded for careless or badly arranged working.

Total Marks 100

Section I: 10 marks

- Attempt Question 1 – 10.
- Answer on the Multiple Choice answer sheet provided.
- Allow about 15 minutes for this section.

Section II: 90 Marks

- Attempt Question 11 - 16
- Answer on blank paper unless otherwise instructed. Start a new page for each new question.
- Allow about 2 hours & 45 minutes for this section.

The answers to all questions are to be returned in separate *stapled* bundles clearly labeled Question 11, Question 12, etc. Each question must show your Candidate Number.

Section I

10 Marks

Attempt Question 1 – 10.

Allow approximately 15 minutes for this section.

Use the multiple choice answer sheet below to record your answers to Question 1 – 10.

Select the alternative: A, B, C or D that best answers the question.

Colour in the response oval completely.

Sample:

$2 + 4 = ?$ (A) 2 (B) 6 (C) 8 (D) 9

A ☐ B ☒ C ☐ D ☐

If you think you have made a mistake, draw a cross through the incorrect answer and colour in the new answer

ie A ☒ B ☒ C ☐ D ☐

If you change your mind and have crossed out what you consider to be the correct answer, then indicate this by writing the word “correct” and draw an arrow as follows:

A ☒ B ☒ C ☐ D ☐
correct

Trial HSC Examination Mathematics Extension 2, 2012

Multiple Choice Answer Sheet

Student id number:

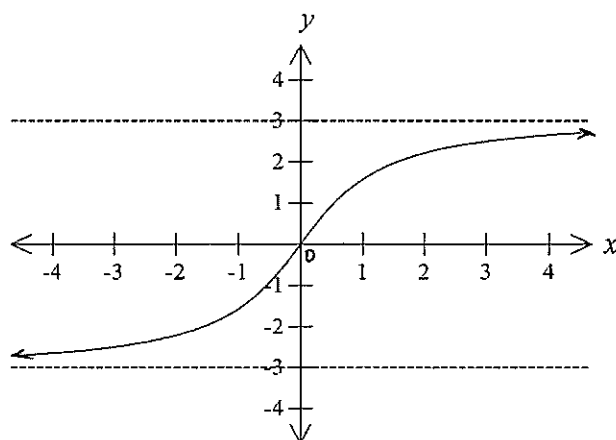
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Completely colour in the response oval representing the most correct answer.

1	A	☐	B	☐	C	☐	D	☐
2	A	☐	B	☐	C	☐	D	☐
3	A	☐	B	☐	C	☐	D	☐
4	A	☐	B	☐	C	☐	D	☐
5	A	☐	B	☐	C	☐	D	☐
6	A	☐	B	☐	C	☐	D	☐
7	A	☐	B	☐	C	☐	D	☐
8	A	☐	B	☐	C	☐	D	☐
9	A	☐	B	☐	C	☐	D	☐
10	A	☐	B	☐	C	☐	D	☐

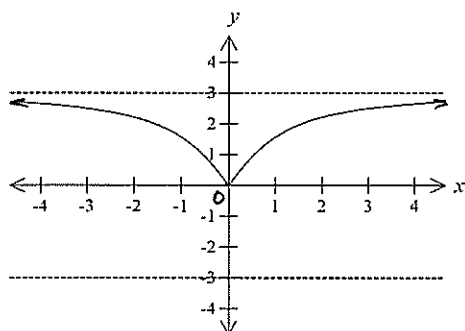
Mark: /10

1. The diagram shows the graph of the function $y = f(x)$.

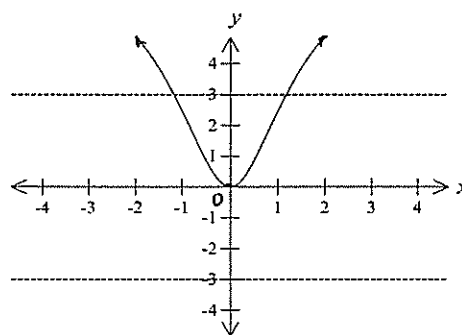


Which of the following is the graph of $y = \sqrt{f(x)}$?

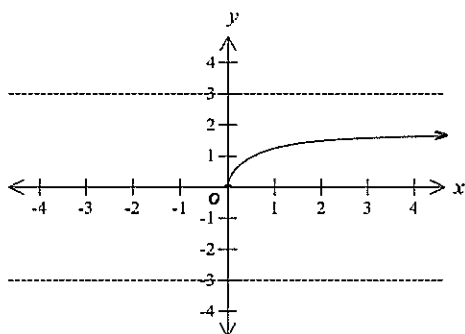
(A)



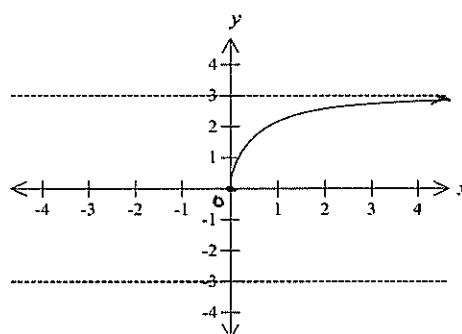
(B)



(C)



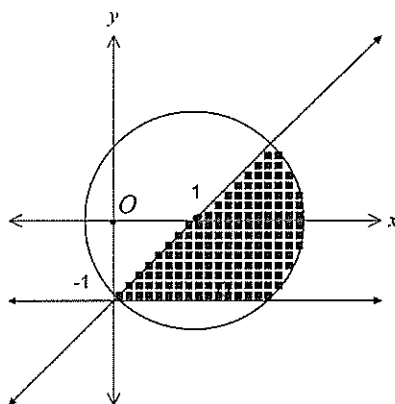
(D)



2. Let $z = 3 - i$. What is the value of $\overline{iz}$?

- (A) $-1 - 3i$.
- (B) $-1 + 3i$.
- (C) $1 - 3i$.
- (D) $1 + 3i$.

3. Consider the Argand diagram below.



Which inequality could define the shaded area?

- (A) $|z - 1| \leq \sqrt{2}$ and $0 \leq \text{Arg}(z - i) \leq \frac{\pi}{4}$.
- (B) $|z - 1| \leq \sqrt{2}$ and $0 \leq \text{Arg}(z + i) \leq \frac{\pi}{4}$.
- (C) $|z - 1| \leq 1$ and $0 \leq \text{Arg}(z - i) \leq \frac{\pi}{4}$.
- (D) $|z - 1| \leq 1$ and $0 \leq \text{Arg}(z + i) \leq \frac{\pi}{4}$.

4. Which of the following is an expression for $\int \frac{1}{\sqrt{x^2 - 6x + 10}} dx$?

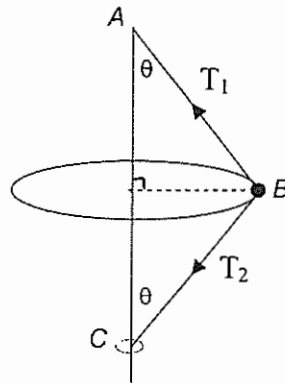
- (A) $\ln(x - 3 - \sqrt{x^2 - 6x + 10}) + c$
- (B) $\ln(x + 3 - \sqrt{x^2 - 6x + 10}) + c$
- (C) $\ln(x - 3 + \sqrt{x^2 - 6x + 10}) + c$
- (D) $\ln(x + 3 + \sqrt{x^2 - 6x + 10}) + c$

5. What is the solution to the inequation: $\frac{x(5-x)}{x-4} \geq -3$?

- (A) $2 \leq x < 4$ or $x \geq 6$.
- (B) $1 \leq x < 4$ or $x \geq 5$.
- (C) $4 < x \leq 6$ or $x \leq 2$.
- (D) $4 > x \leq 5$ or $x \leq 1$.

6. The points $P(a \cos \theta, b \sin \theta)$ and $Q(a \cos \phi, b \sin \phi)$ lie on the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ and the chord PQ subtends a right angle at $(0,0)$. Which of the following is the correct expression?
- (A) $\tan \theta \tan \phi = -\frac{b^2}{a^2}$.
- (B) $\tan \theta \tan \phi = -\frac{a^2}{b^2}$.
- (C) $\tan \theta \tan \phi = \frac{b^2}{a^2}$.
- (D) $\tan \theta \tan \phi = \frac{a^2}{b^2}$.
7. What are the values of real numbers p and q such that $1-i$ is a root of the equation $z^3 + pz + q = 0$?
- (A) $p = 2$ and $q = 4$.
- (B) $p = 2$ and $q = -4$.
- (C) $p = -2$ and $q = 4$.
- (D) $p = -2$ and $q = -4$.
8. A particle of mass m is projected vertically upwards with an initial velocity of $u \text{ ms}^{-1}$ in a medium in which the resistance to the motion is proportional to the square of the velocity $v \text{ ms}^{-1}$ of the particle or mkv^2 . Let x be the displacement in metres of the particle above the point of projection, O , so that the equation of motion is $\ddot{x} = -(g + kv^2)$ where $g \text{ ms}^{-2}$ is the acceleration due to gravity. Assume $k = 10$ and the acceleration due to gravity is 10 ms^{-2} . Which of the following gives the correct expression for the time taken?
- (A) $t = \frac{1}{10}(\tan^{-1} u - \tan^{-1} v)$.
- (B) $t = \frac{1}{10}(\tan^{-1} v - \tan^{-1} u)$.
- (C) $t = \frac{1}{10}(\tan^{-1} u + \tan^{-1} v)$.
- (D) $t = \frac{1}{10}(\tan^{-1} v + \tan^{-1} u)$.

9. A body of mass m kg is attached by two light rods AB and BC . Both rods are l metres in length. Rod AB is hinged at point A and makes an angle θ with the vertical shaft. Rod BC is attached to a ring that can slide freely along the vertical shaft.



What are the tensions in the rods?

- (A) $T_1 = \frac{1}{2}(mg \sec \theta + ml\omega^2)$ and $T_2 = \frac{1}{2}(ml\omega^2 - mg \sec \theta)$.
- (B) $T_1 = \frac{1}{2}(mg \sin \theta + ml\omega^2)$ and $T_2 = \frac{1}{2}(ml\omega^2 - mg \sin \theta)$.
- (C) $T_1 = \frac{1}{2}(mg \sec \theta - ml\omega^2)$ and $T_2 = \frac{1}{2}(ml\omega^2 + mg \sec \theta)$.
- (D) $T_1 = \frac{1}{2}(mg \sin \theta - ml\omega^2)$ and $T_2 = \frac{1}{2}(ml\omega^2 + mg \sin \theta)$.
10. A skydiver falls from a plane which is flying horizontally at 2 000 m. Initially his motion is determined by the acceleration due to gravity of 10 m/s^2 and any resistance is negligible. After 5 seconds, he opens his parachute and his motion is determined by the equation: $\ddot{x} = 10 - \frac{5}{4}v$, where downwards direction is taken as positive.
- Hence his terminal velocity will be 8 m/s.
- Which statement best reflects the situation after the skydiver opens his parachute?
- (A) He hits the ground with a vertical speed of 50 m/s.
- (B) He hits the ground with a vertical speed of 8 m/s
- (C) His vertical speed never exceeds 8 m/s.
- (D) His vertical speed never drops below 8 m/s.

End of Section I

Section II

Total Marks is 90

Attempt Question 11 – 16.

Allow approximately 2 hours & 45 minutes for this section.

Answer all questions, starting each new question on a new sheet of paper with your **student id number** in the top right hand corner and the question number on the left hand side of your paper.

All necessary working must be shown in each and every question.

Question 11. [Start a New Page]

Marks

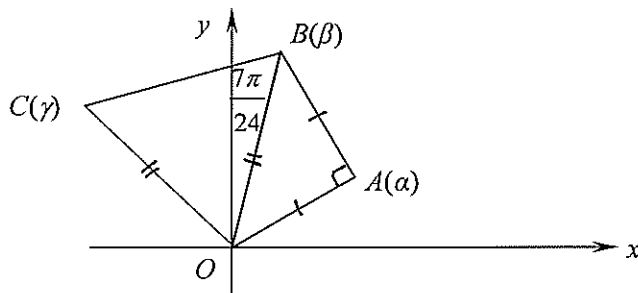
(a) Given $z = 1 + 2i$ and $w = -2 + i$, find:

- | | | |
|-------|------------------|---|
| (i) | $ z $. | 1 |
| (ii) | zw . | 2 |
| (iii) | $\frac{5}{iw}$. | 2 |

(b) Find the two complex numbers z that satisfy: $z\bar{z} = 37$ and $\frac{z}{\bar{z}} = \frac{35}{37} + \frac{12i}{37}$. 3

(c) If $w = (-1 + i\sqrt{3})^{2012}$, find $\text{Arg } w$. 2

(d) Points A, B and C represent the complex numbers α, β and γ in the Argand diagram respectively.



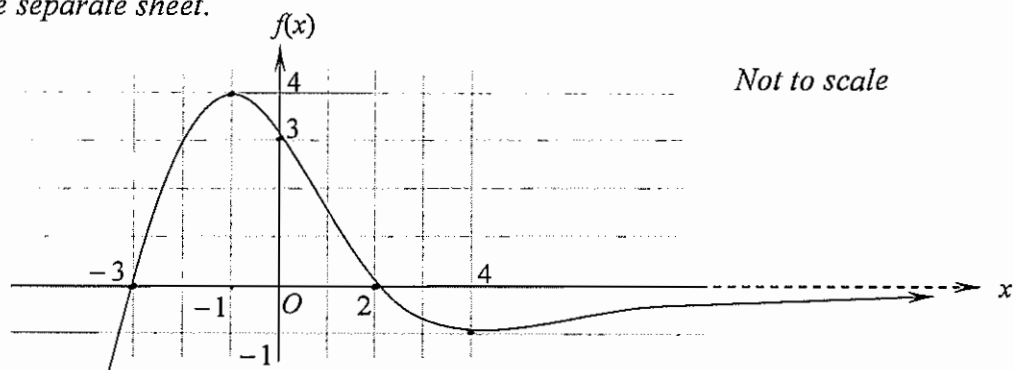
$\triangle OAB$ is right isosceles at A , $\triangle COB$ is isosceles with $OB = OC$ and $\angle OBC = \frac{7\pi}{24}$.

(i) Copy or trace the diagram onto your writing and find $\angle AOC$. 1

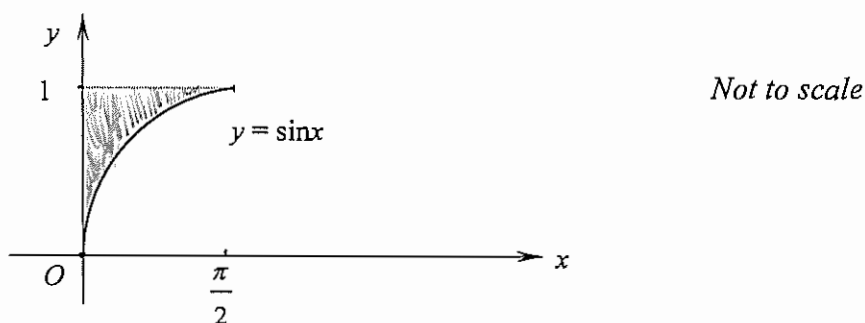
(ii) Explain why $\gamma = \sqrt{2} \left(\cos \frac{2\pi}{3} + i \sin \frac{2\pi}{3} \right) \alpha$. 2

(iii) Hence find the value of $2\alpha^2 + \gamma^2 + \alpha\gamma\sqrt{2}$. 2

- (a) Given the sketch of the function $f(x)$, sketch each of the following on separate diagrams:
See separate sheet.



- (i) $y = -f(x)$. 1
- (ii) $y = f(-x)$. 2
- (iii) $y = f(x^2)$. 2
- (b) (i) Show that the equation of the tangent to the polynomial function $y = P(x)$ at $x = \alpha$ is $y = P'(\alpha)(x - \alpha) + P(\alpha)$. 1
- (ii) Explain why: when the polynomial $P(x)$ (of degree greater than 2) is divided by the quadratic $(x - \alpha)^2$, then $P(x) = (x - \alpha)^2 Q(x) + ax + b$, where $Q(x)$ is the quotient and a and b are real numbers. 1
- (iii) Hence show that when $P(x)$ is divided by $(x - \alpha)^2$, the remainder is equivalent to $P'(\alpha)(x - \alpha) + P(\alpha)$, the expression from the tangent in part (b) (i). 3
- (c) The area bounded by the curve $y = \sin x$, for $0 \leq x \leq \frac{\pi}{2}$, the lines $x = 0$ and $y = 1$ is rotated about the y -axis.

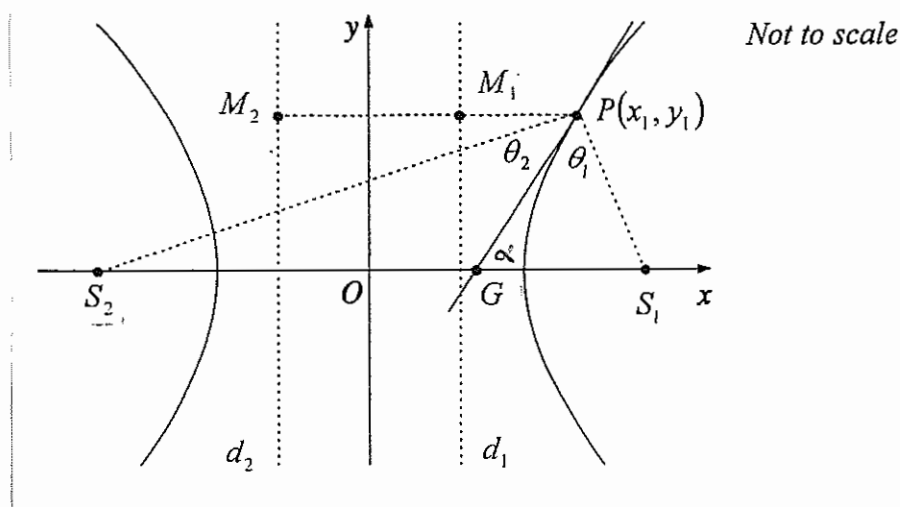


- (i) By using the method of cylindrical shells, show that the volume V of the solid of revolution about the y -axis is given by:

$$V = 2\pi \int_0^{\frac{\pi}{2}} x(1 - \sin x) dx.$$

- (ii) Hence calculate the volume of the solid. 3

(a)



The point $P(x_1, y_1)$ lies on the hyperbola $\frac{x^2}{25} - \frac{y^2}{9} = 1$.

The two foci of the hyperbola are S_1 and S_2 and the two directrices are d_1 and d_2 , as shown.

- (i) Show that the length $S_1P = \frac{\sqrt{34}}{5}x_1 - 5$. 2
- (ii) Show that the equation of the tangent at P is $\frac{x_1x}{25} - \frac{y_1y}{9} = 1$. 2
- (iii) The tangent at P intersects the transverse axis at point G .
Find the coordinates of point G . 1
- (iv) Given $\angle S_1PG = \theta_1$, $\angle GPS_2 = \theta_2$ and $\angle S_1GP = \alpha$,
 - (1) By using the sine rule, show that: $\sin \alpha = \frac{x_1}{5} \sin \theta_1$. 2
 - (2) Hence show that: $\sin \theta_1 = \sin \theta_2$. 2
 - (3) Hence deduce that GP bisects $\angle S_1PS_2$. 2

(b) Given that $I_n = \int_{\frac{\pi}{6}}^{\frac{\pi}{4}} \cot^n x \, dx$, for $n = 1, 2, \dots$

- (i) Show that $I_1 = \frac{1}{2} \ln 2$. 1
- (ii) Show that: $I_{n-2} + I_n = \frac{1}{n-1} \left(3^{\frac{1}{2}(n-1)} - 1 \right)$, for $n = 2, 3, 4, \dots$ 2
- (iii) Find I_5 . 1

- (a) Given that: $\theta_1 + \theta_2 + \theta_3 + \dots + \theta_n < \frac{\pi}{2}$, where $0 \leq \theta_i < \frac{\pi}{2}$ for $i = 1, 2, 3, \dots, n$. 3
 Prove, by mathematical induction for $n = 1, 2, 3, \dots$, that:

$$\tan(\theta_1 + \theta_2 + \theta_3 + \dots + \theta_n) \geq \tan \theta_1 + \tan \theta_2 + \tan \theta_3 + \dots + \tan \theta_n.$$

- (b) (i) Find the constants a , b and c such that: 3

$$\frac{300x}{1000 + x^3} = \frac{a}{10 + x} + \frac{bx + c}{100 - 10x + x^2}.$$

- (ii) A particle of mass m kg is projected vertically upwards in a highly resistive medium at a velocity of 5 m/s.
 The particle is subjected to the force of gravity and to a resistance due to the medium of magnitude $\frac{mv^3}{100}$ newtons. Given the acceleration due to gravity is 10 m/s^2 ,

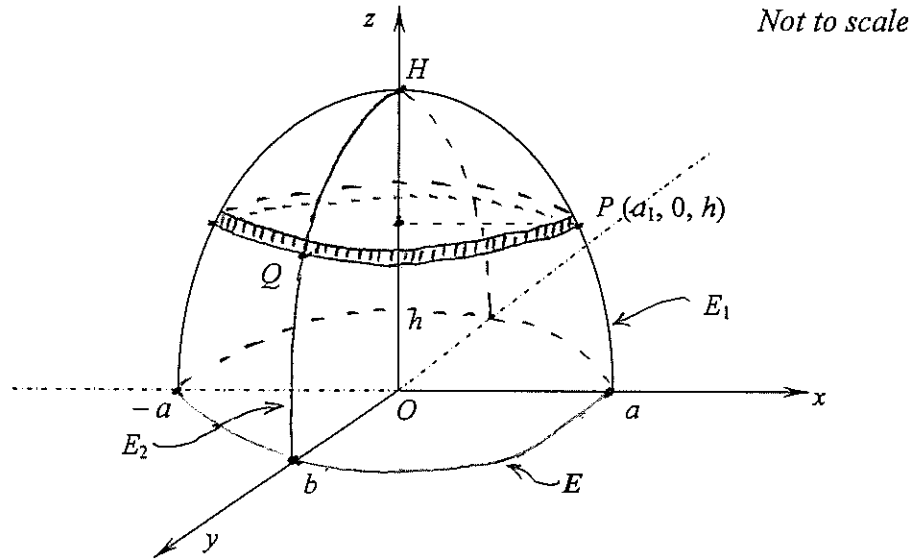
- (1) State the equation of motion (if upwards is the positive direction) 1
 (2) Hence find the maximum height reached by the particle, (giving your answer correct to 1 decimal places). 3

Question 14 (c) continued over page

- (c) A right solid has an elliptical base whose equation is $E: \frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$.

The height of the solid is H such that the sections in the x - z plane and the y - z plane are the semi-ellipses E_1 and E_2 respectively.

Every cross-section parallel to the base is elliptical in shape as shown in the diagram.



- (i) Given that the equation of ellipse E_1 is $\frac{x^2}{a^2} + \frac{z^2}{H^2} = 1$, 1
 State the equation for ellipse E_2 .
- (ii) By taking a slice parallel to the base at height of h of thickness Δh , 2
 Hence point P can be stated as $(a_1, 0, h)$ and Q as $(0, b_1, h)$ for the elliptical slice,
 as shown in the diagram.

By assuming that the area of ellipse E is πab square units,

Show that the cross-sectional area A of this slice is given by $A = \pi ab \left(1 - \frac{h^2}{H^2} \right)$.

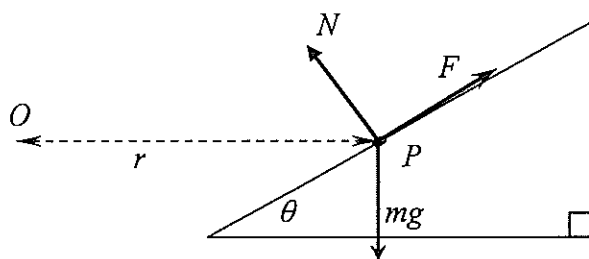
- (iii) Hence find the volume of the solid. 2

Question 15. [Start a New Page]

Marks

- (a) Let α, β and γ be the distinct roots of the cubic equation $x^3 + ax^2 + bx - 54 = 0$, where a and b are real numbers. Suppose that $\alpha^2 + \beta^2 = 0$ and $\alpha^2 + \gamma^2 = 0$,
- (i) Explain why $\beta + \gamma = 0$. 2
 - (ii) Hence explain why α is real. 2
 - (iii) Hence, or otherwise explain why β and γ are complex and purely imaginary. 2
 - (iv) Find a and b . 2

- (b) On a racetrack for small cars of mass m kg, a circular bend of radius r m is banked at an angle of θ to the horizontal. The maximum frictional force is F Newtons (up or down the bank) and the acceleration due to gravity is 10 m/s^2 ie $g = 10$. The normal reaction to the surface is N Newtons. Let point P represent the small car on the banked track as shown in the diagram.



Not to scale

- (i) From the diagram, the vertical resolution for motion downwards at P is: 1

$$N \cos \theta + F \sin \theta = mg.$$

Find the horizontal resolution when the car is travelling at speed v m/s at P .
- (ii) Hence, if $r = 80$, $\theta = 45^\circ$ and the maximum frictional force is at most $\frac{1}{9}$ of the 2
 normal reaction force N . Find the minimum speed that the car can safely negotiate the bend without slipping down the incline.
- (iii) For the upwards motion of the car, find the maximum speed that the car can 3
 safely negotiate the bend.
- (iv) Hence or otherwise, determine the designed speed (no slipping) for this angle. 1

Question 16. [Start a New Page]

Marks

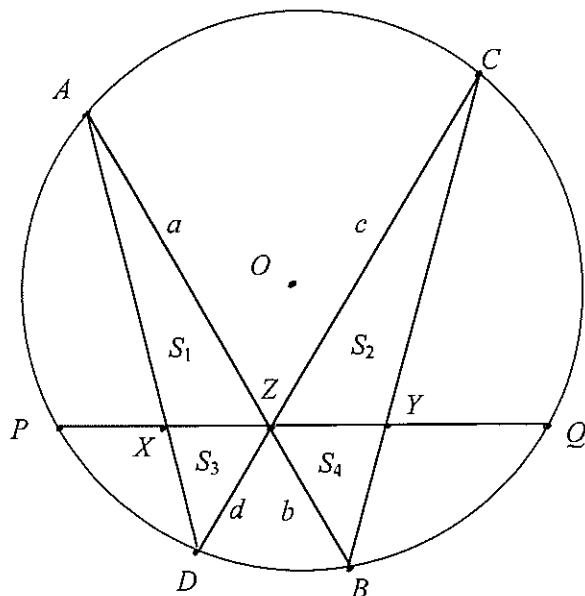
(a) Show that $\int_0^1 \frac{\ln(1+x)}{1+x^2} dx = \frac{\pi}{8} \ln 2$, using the substitution $x = \frac{1-u}{1+u}$.

4

(b) The roots of $x^4 + 3x - 1 = 0$ are α, β, γ and δ . Find $\alpha^4 + \beta^4 + \gamma^4 + \delta^4$.

2

(c)



Not to scale

Given Z is any point on the chord PQ of the given circle. Chord AB and CD pass through Z . Let X and Y be the points of intersection of the chords AD and CB with PQ respectively, as shown in the diagram.

Given $ZA = a$; $ZB = b$; $ZC = c$ and $ZD = d$. **Let** $ZP = p$; $ZQ = q$; $ZX = x$ and $ZY = y$.

Given S_1 denotes the *area* of $\triangle ZAX$; S_2 for $\triangle ZCY$; S_3 for $\triangle ZDX$ and S_4 for $\triangle ZBY$.

(i) By detaching page **12** and stapling it to your Question 16,

Show that: $\frac{S_1}{S_2} = \frac{a \cdot AX}{c \cdot CY}$ and $\frac{S_1}{S_4} = \frac{a \cdot x}{b \cdot y}$.

3

(ii) Hence deduce that: $\frac{S_1 S_3}{S_2 S_4} = \frac{a \cdot d \times AX \cdot XD}{b \cdot c \times CY \cdot YB} = \frac{a \cdot d \cdot x^2}{b \cdot c \cdot y^2}$.

2

(iii) Hence explain why: $\frac{x^2}{y^2} = \frac{(p-x)(x+q)}{(p+y)(q-y)}$.

1

(iv) Hence show that: $\frac{1}{x} - \frac{1}{y} = \frac{1}{p} - \frac{1}{q}$.

2

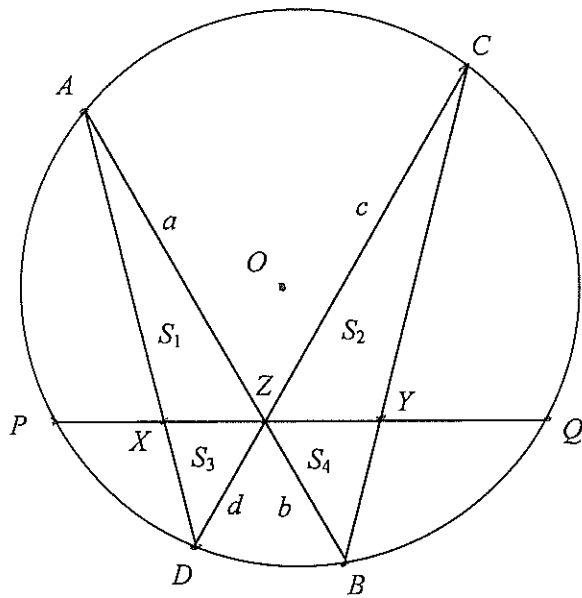
(v) If Z is the midpoint of PQ , what is the relationship between x and y .

1

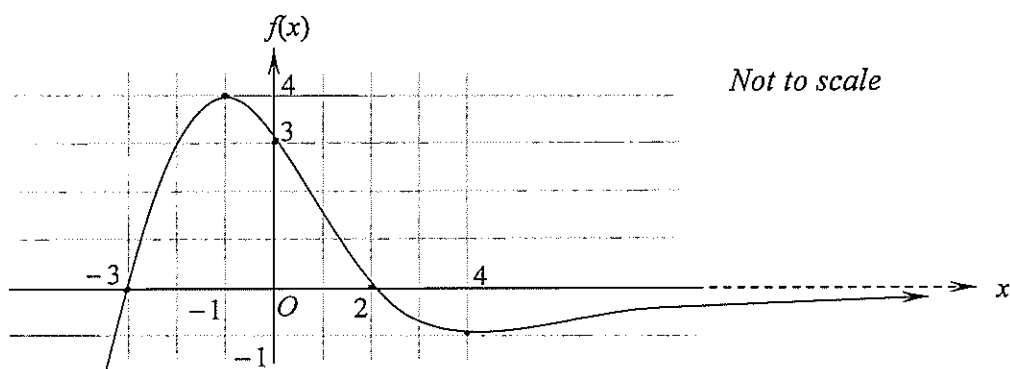
THE END



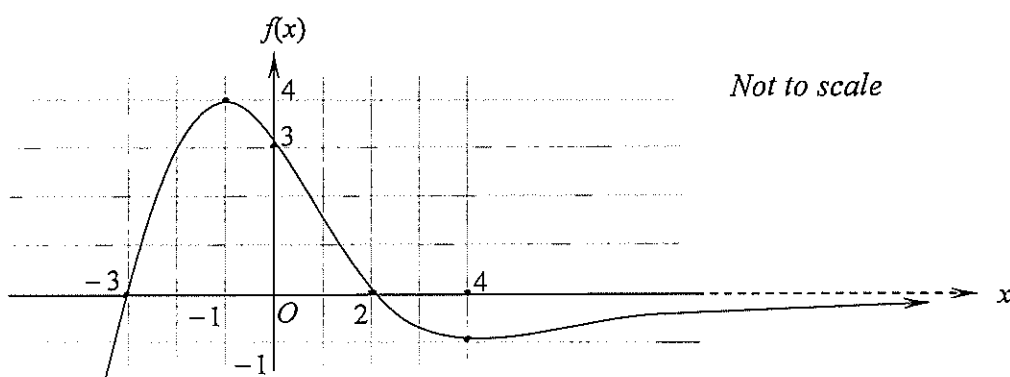
Question 16. (c) Attach this diagram to your **Q16 (c)** solutions.



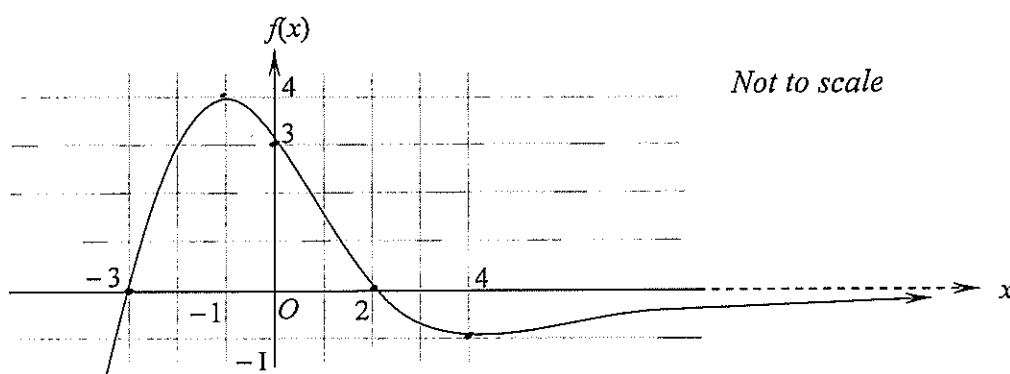
(i)



(ii)



(iii)



Section I

10 Marks

Attempt Question 1 – 10.

Allow approximately 15 minutes for this section.

Use the multiple choice answer sheet below to record your answers to Question 1 – 10.

Select the alternative: A, B, C or D that best answers the question.

Colour in the response oval completely.

Sample:

$2 + 4 = ?$ (A) 2 (B) 6 (C) 8 (D) 9

A ☐ B ☒ C ☐ D ☐

If you think you have made a mistake, draw a cross through the incorrect answer and colour in the new answer

ie A ☒ B ☒ C ☐ D ☐

If you change your mind and have crossed out what you consider to be the correct answer, then indicate this by writing the word “correct” and draw an arrow as follows:

A ☒ B ☒ C ☐ D ☐
 correct
 A ☒ B ☒ C ☐ D ☐

Trial HSC Examination Mathematics Extension 2, 2012

Multiple Choice Answer Sheet

Student id number:

SOLUTIONS

ANSWERS.

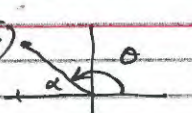
Completely colour in the response oval representing the most correct answer.

1	A	☐	B	☐	C	☒	D	☐
2	A	☐	B	☐	C	☒	D	☐
3	A	☐	B	☒	C	☐	D	☐
4	A	☐	B	☐	C	☒	D	☐
5	A	☐	B	☐	C	☒	D	☐
6	A	☐	B	☒	C	☐	D	☐
7	A	☐	B	☐	C	☒	D	☐
8	A	☒	B	☐	C	☐	D	☐
9	A	☒	B	☐	C	☐	D	☐
10	A	☐	B	☐	C	☐	D	☒

Mark: /10

TRAHS M. EXT 2 TRIAL, 2012 SECTION II SOLUTIONS

MATHEMATICS Extension 2: Question.....

Suggested Solutions	Marks	Marker's Comments
Q11. (a) (i) $z = 1 + 2i$, $w = -2 + i$ $z = \sqrt{1^2 + 2^2}$ $\therefore z = \sqrt{5}$		11
(ii) $zw = (1 + 2i)(-2 + i)$ $= -2 + i - 4i + 2i^2 = -2 + i - 4i - 2$ $\therefore zw = -4 - 3i$		2
(iii) $\frac{5}{iw} = \frac{5}{i(-2 + i)} = \frac{5}{-1 - 2i} \times \frac{-1 + 2i}{-1 + 2i}$ $= \frac{5(-1 + 2i)}{(-1)^2 + 4} = \frac{5(-1 + 2i)}{5}$ $\therefore \frac{5}{iw} = -1 + 2i$		2 $\frac{5}{iw} \times \frac{iw}{iw} = \frac{-5iw}{ w ^2}$ $= \frac{-5i(-2 + i)}{5}$
(b) $z\bar{z} = 37$ $\frac{z}{\bar{z}} = \frac{35 + 12i}{37}$ Now $\frac{z}{\bar{z}} \times \frac{z}{\bar{z}} = \frac{z^2}{z\bar{z}} = \frac{z^2}{37}$ $\therefore \frac{z^2}{37} = \frac{35 + 12i}{37} \Rightarrow z^2 = 35 + 12i$ Let $z = x + iy$ $x, y \in \mathbb{R}$ $z^2 = x^2 - y^2 + 2xyi$ $z\bar{z} = x^2 + y^2$ $\therefore x^2 - y^2 = 35$ — (1) $2xy = 12$ $xy = 6$ — (2) $x^2 + y^2 = 37$ — (3) (1) and (3) $2x^2 = 72$ $x^2 = 36$ $x = \pm 6$ sub in (2) $y = \frac{6}{x} = 1 \text{ or } -1$ $\therefore z = 6 + i \text{ or } -6 - i$ $\pm(6 + i)$		3 $z = x + iy$ $\bar{z} = x - iy$ $z\bar{z} = x^2 + y^2$ $z^2 = x^2 - y^2 + 2xyi$
(c) $w = (-1 + i\sqrt{3})^{2012}$ $(-1, \sqrt{3})$  $\tan \alpha = \frac{\sqrt{3}}{-1} \Rightarrow \alpha = \frac{2\pi}{3}$ $\therefore \theta = \frac{2\pi}{3}$ $\text{Arg } w = 2012 \text{ Arg } (-1 + i\sqrt{3})$ $= 2012 \times \frac{2\pi}{3} = \frac{4024\pi}{3} \quad (\div 2\pi)$ $= -\frac{2\pi}{3}$		2012 3 $= \frac{\pi}{3}$ 2

JRAHS M. EXT 2 TRIAL, 2012 SECTION II SOLUTIONS

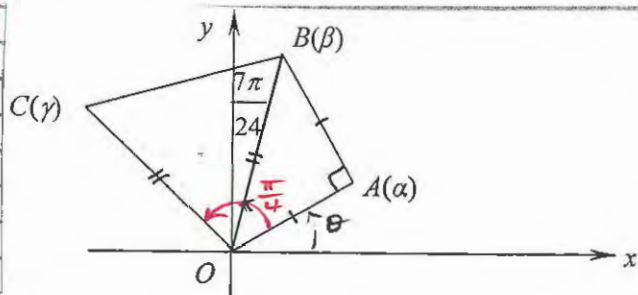
MATHEMATICS Extension 2: Question 11

Suggested Solutions

Marks

Marker's Comments

(d)



(i)

$$\angle AOB = \frac{\pi}{4} \quad \text{right isosceles } \triangle OAB$$

$$\angle OCB = \frac{7\pi}{24} \quad (\text{equal angles opposite equal sides } OB=OC)$$

$$2 \times \frac{7\pi}{24} + \angle COB = \pi \quad (\text{angle sum of } \triangle COB \text{ is } \pi)$$

$$\angle COB = \pi - \frac{7\pi}{12} = \frac{5\pi}{12}$$

$$\therefore \angle AOC = \frac{\pi}{4} + \frac{5\pi}{12} = \frac{8\pi}{12} = \frac{2\pi}{3}$$

1

(ii)

$$\vec{OC} = \gamma, \quad \text{Arg } \gamma = \theta + \angle ABC = \theta + \frac{2\pi}{3}, \quad \text{Arg } \alpha = \theta$$

$$OC = OB = |\beta| = \sqrt{2} \cdot OA$$

$$\text{but } OC = |\gamma| = |\beta| = \sqrt{2} \cdot OA$$

(right isosceles $\triangle$, ...)
Pyth. Thm)

Enlarge by
cancel
Rotate by

$$\therefore \gamma = \beta \operatorname{cis} \frac{2\pi}{3} = \sqrt{2} \cdot \alpha \cdot \operatorname{cis} \frac{2\pi}{3} = \sqrt{2} (\cos \frac{2\pi}{3} + i \sin \frac{2\pi}{3}) \alpha$$

2

(iii)

$$\text{LHS} = 2\alpha^2 + \gamma^2 + \alpha\gamma\sqrt{2} = 2\alpha^2 + [\sqrt{2} \operatorname{cis} \frac{2\pi}{3} \alpha]^2 + \alpha \cdot \sqrt{2} \operatorname{cis} \frac{2\pi}{3} \cdot \alpha \cdot \sqrt{2}$$

$$= 2\alpha^2 + 2\alpha^2 \operatorname{cis} \frac{4\pi}{3} + 2\alpha^2 \operatorname{cis} \frac{2\pi}{3}$$

(de Moivre's Thm)

$$= 2\alpha^2 [1 + \operatorname{cis} \frac{4\pi}{3} + \operatorname{cis} \frac{2\pi}{3}]$$

$$= 2\alpha^2 [1 + -\frac{1}{2} - \frac{\sqrt{3}}{2}i + -\frac{1}{2} + \frac{\sqrt{3}}{2}i]$$

$$= 0$$

2

TRAHS M. EXT 2 TRIAL, 2012 SECTION II SOLUTIONS

MATHEMATICS Extension 2: Question 12

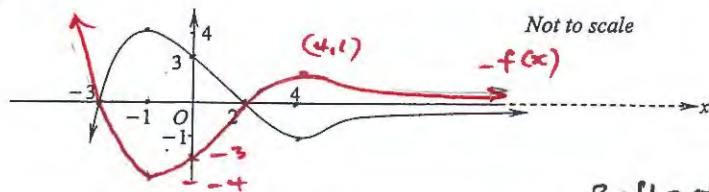
Suggested Solutions

Marks

Marker's Comments

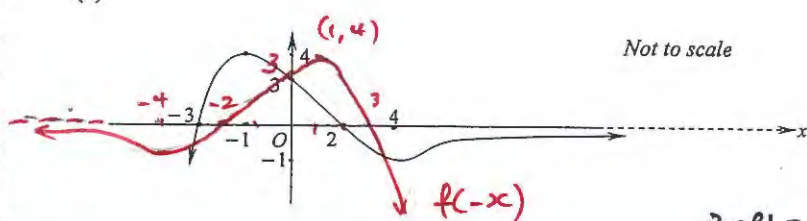
Question 12 (a)

(i)



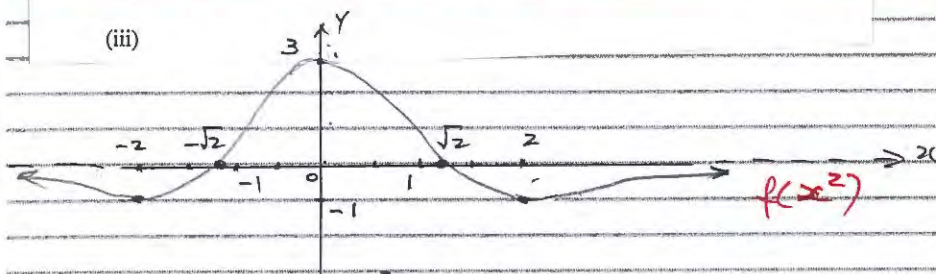
Reflection in x-axis

(ii)



Reflection in y-axis.

(iii)



New x-int: $x^2 = -3$; -1 ; 2
 $\therefore x = \pm\sqrt{-3}$; ± 1 ; $\pm\sqrt{2}$

$x=0$ $x^2=0$ $f(x^2)=f(0)=3$ $(0, 3)$
 $x=\pm 2$ $x^2=4$ $f(x^2)=f(4)=-1$ $(\pm 2, -1)$

(b) (i) $y = P(x) \therefore \frac{dy}{dx} = P'(x)$

Gradient of tangent at $(\alpha, P(\alpha))$ $m_T = P'(\alpha)$

Eqn of Tangent: $y - P(\alpha) = P'(\alpha)(x - \alpha)$

ie $y = P'(\alpha)(x - \alpha) + P(\alpha)$

QED

(ii) $P(x) = (x - \alpha)^2 Q(x) + R(x)$

Since the divisor $(x - \alpha)^2$ is of degree 2

$\therefore$ deg of remainder $R(x)$ is at most 1

$\therefore R(x)$ is linear

ie $R(x) = ax + b$

MATHEMATICS Extension 2: Question 12

Suggested Solutions

Marks

Marker's Comments

(b)(iii) $P(x) = (x-a)^2 Q(x) + ax + b$

$\therefore P(a) = 0 + aa + b$

$P(a) = aa + b \quad \text{--- (1)}$

$P'(x) = 2(x-a)Q(x) + (x-a)^2 Q'(x) + a$

$\therefore P'(a) = 0 + 0 + a$

$\therefore a = P'(a) \quad \text{--- (2)}$

So $R(x) = ax + b$

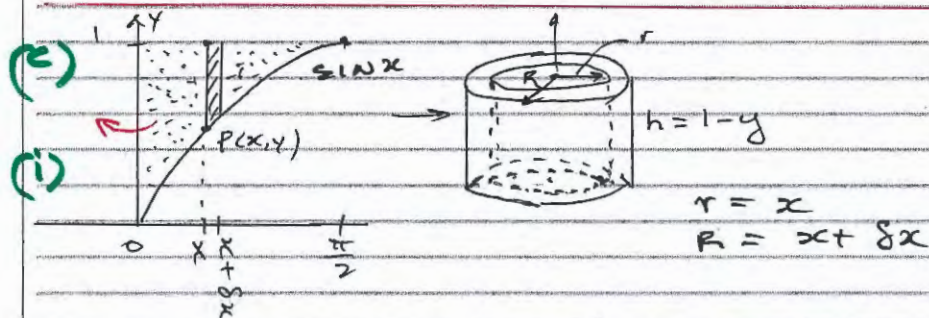
$= P'(a)x + P(a) - ax$

$= P'(a)x - P'(a)a + P(a)$

$= P'(a)(x-a) + P(a)$

$\therefore$ Remainder is the same as the tangent at $x=a$ when divide by $(x-a)^2$.

3



" $r = x - \frac{1}{2} \delta x$ "

" $R = x + \frac{1}{2} \delta x$ "

" $A \approx \pi \times 2x \times \delta x$ "

"Rectangular Prism"

" $A \approx 2\pi x(1-y)$ "

" $\delta V \approx 2\pi x(1-y) \delta x$ "

$A \approx \pi [R^2 - r^2] = \pi [(R+r)(R-r)]$

$A \approx \pi [(2x + \delta x)(\delta x)] = \pi [2x\delta x + (\delta x)^2]$

$A \approx 2\pi x \delta x$, neglect $(\delta x)^2 \approx 0$

$\therefore$ Volume of shell $\delta V \approx 2\pi x \delta x (1-y) = 2\pi x(1-y) \delta x$

Volume of solid $V \approx \sum_{x=0}^{\pi/2} 2\pi x(1-\sin x) \delta x$

$= \lim_{\delta x \rightarrow 0} \sum_{x=0}^{\pi/2} 2\pi x(1-\sin x) \delta x$

$= 2\pi \int_0^{\pi/2} x(1-\sin x) dx$

2

(ii) $V = 2\pi \int_0^{\pi/2} x(1-\sin x) dx$ $u = x$ $dv = (1-\sin x) dx$ METHOD I

$= 2\pi \left\{ \left[x(x + \cos x) \right]_0^{\pi/2} - \int_0^{\pi/2} (x + \cos x) dx \right\}$

$= 2\pi \left\{ \left[\frac{\pi}{2} \cdot \frac{\pi}{2} - 0 \right] - \left[\frac{x^2}{2} + \sin x \right]_0^{\pi/2} \right\}$

$= 2\pi \left\{ \frac{\pi^2}{4} - \left(\frac{\pi^2}{8} + 1 \right) \right\}$

$= 2\pi \left\{ \frac{\pi^2}{4} - \frac{\pi^2}{8} - 1 \right\} = 2\pi \left\{ \frac{\pi^2}{8} - 1 \right\}$

Volume is $\left(\frac{\pi^3}{4} - 2\pi \right) u^3$

$V = \left[x + \cos x \right]_0^{\pi/2}$
 $2\pi \left[\frac{x^2}{2} + x \cos x - \sin x \right]_0^{\pi/2}$

3

TRANS M. EXT 2 TRIAL, 2012 SECTION II SOLUTIONS

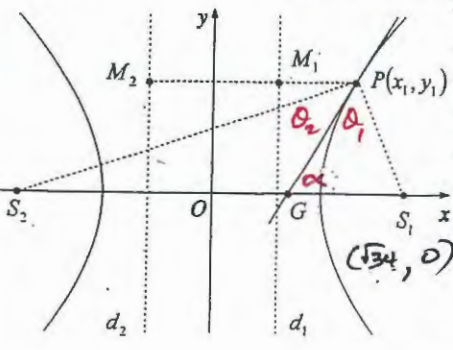
MATHEMATICS Extension 2: Question.....13

Suggested Solutions

Marks

Marker's Comments

(i)



$$\frac{x^2}{25} - \frac{y^2}{9} = 1 \quad a=5, b=3$$

$$e^2 = 1 + \frac{b^2}{a^2} = 1 + \frac{9}{25} = \frac{34}{25}$$

$$e = \frac{\sqrt{34}}{5}; e > 1$$

$$S_1(ae, 0) = (\sqrt{34}, 0)$$

$$d_1: x = \frac{a}{e} = \frac{25}{\sqrt{34}}$$

$\frac{34}{25}$

Distance formula
 $S_1P = \sqrt{(\sqrt{34} - x_1)^2 + (0 - y_1)^2}$

2

2

(ii)

$$\frac{x^2}{25} - \frac{y^2}{9} = 1$$

$$\frac{d}{dx} \left[\frac{x^2}{25} - \frac{y^2}{9} \right] = \frac{d}{dx} 1$$

$$\frac{2x}{25} - \frac{2y}{9} y' = 0$$

$$\frac{dy}{dx} = \frac{9x}{25y}$$

Gradient of tangent at P: $m_T = \frac{9x_1}{25y_1}$

Eqn. of Tangent $y - y_1 = \frac{9x_1}{25y_1} (x - x_1)$

$$\frac{x_1 y_1}{9} \quad \frac{y_1 y - y_1^2}{9} = \frac{x_1 x - x_1^2}{25}$$

$$\text{So } \frac{x_1 x}{25} - \frac{y_1 y}{9} = \frac{x_1^2}{25} - \frac{y_1^2}{9}$$

$$\text{ie } \frac{x_1 x}{25} - \frac{y_1 y}{9} = 1 \quad \text{as } \frac{x_1^2}{25} - \frac{y_1^2}{9} = 1 \quad \text{qed}$$

as $P(x_1, y_1)$ is on Hyp.

(iii)

For G $y=0 \therefore \frac{x_1 x}{25} = 1$

$$\text{ie } x = \frac{25}{x_1}$$

$$\text{ie } Q = \left(\frac{25}{x_1}, 0 \right)$$

1

(iv)

For ΔS_1PG : $S_1P = S_1Q$

$$\frac{\sin \alpha}{\sin \theta_1} = \frac{\sin \theta_2}{\sin \theta_1}$$

$$\sin \alpha = \frac{S_1P \sin \theta_1}{S_1Q} = \frac{(\frac{\sqrt{34}}{5} x_1 - 5) \sin \theta_1}{\sqrt{34} - \frac{25}{x_1}}$$

$$= \frac{(\sqrt{34} x_1 - 25) x_1 \sin \theta_1}{(\sqrt{34} x_1 - 25) 5}$$

$$\therefore \sin \alpha = \frac{x_1}{5} \sin \theta_1 \quad \text{qed.}$$

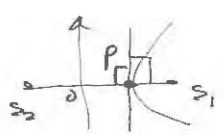
$$S_1Q = \sqrt{34} - \frac{25}{x_1}$$

$$x_1 \neq \frac{25}{\sqrt{34}}$$

2

TRAHS M. EXT 2 TRIAL, 2012 SECTION II SOLUTIONS

MATHEMATICS Extension 2: Question 13

Suggested Solutions	Marks	Marker's Comments
(iv) (2) In $\triangle S_2PG$, $\angle PGS_2 = \pi - \alpha$ $\frac{S_2P}{\sin(\pi - \alpha)} = \frac{S_2G}{\sin \theta_2}$ $\frac{S_2P}{\sin \alpha} = \frac{S_2G}{\sin \theta_2}$ $\sin \alpha = \frac{S_2P \sin \theta_2}{S_2G}$ $= \frac{x_1 \sin \theta_2}{5}$ so $\sin \alpha = \frac{x_1 \sin \theta_1}{5} = \frac{x_1 \sin \theta_2}{5}$ $\therefore \sin \theta_1 = \sin \theta_2$	$S_2P = GP \times 2 = \frac{\sqrt{34}x_1 + 5}{5} = \frac{\sqrt{34}x_1 + 25}{5}$ $S_2G = \frac{25}{x_1} + \sqrt{34} = \frac{\sqrt{34}x_1 + 25}{x_1}$	[2]
(3) $\sin \theta_1 = \sin \theta_2$ $\therefore \theta_1 = \theta_2 \text{ or } \theta_1 = \pi - \theta_2$ Now $\theta_1 = \pi - \theta_2$ only when $\theta_1 = \theta_2 = \frac{\pi}{2}$ ie. P is at vertex A $\therefore \theta_1 = \theta_2$ $\therefore GP \text{ bisects } \angle S_1PS_2$		[2]
(b) $I_n = \int_{\pi/6}^{\pi/4} \cot^n x \, dx \quad n = 1, 2, 3, \dots$ (i) $I_1 = \int_{\pi/6}^{\pi/4} \frac{\cos x}{\sin x} \, dx = \ln(\sin x) \Big _{\pi/6}^{\pi/4}$ $= \ln \frac{1}{\sqrt{2}} - \ln \frac{1}{2} = \ln \frac{2}{\sqrt{2}} = \ln \sqrt{2} = \ln 2^{\frac{1}{2}}$ $I_1 = \frac{1}{2} \ln 2 \quad \text{QED}$	$\sin x > 0$ $[\frac{\pi}{6}, \frac{\pi}{4}]$	[1]
(ii) $I_{n-2} + I_n = \int_{\pi/6}^{\pi/4} \cot^{n-2} x + \cot^n x \, dx$ $= \int_{\pi/6}^{\pi/4} \cot^{n-2} x (1 + \cot^2 x) \, dx$ $= \int_{\pi/6}^{\pi/4} \cot^{n-2} x \cdot \operatorname{cosec}^2 x \, dx$ $= -\frac{1}{n-1} \cot^{n-1} x \Big _{\pi/6}^{\pi/4}$ $= -\frac{1}{n-1} [1 - (\sqrt{3})^{n-1}] = \frac{1}{n-1} [3^{\frac{1}{2}(n-1)} - 1]$		[2]
(iii) I_5: $n=5 \quad I_3 + I_5 = \frac{1}{4} [3^2 - 1] = 2$ $n=3 \quad I_1 + I_3 = \frac{1}{2} [3 - 1] = 1$ $\therefore I_5 = 2 - I_3 = 2 - (1 - I_1) = 1 + I_1$ $= 1 + \frac{1}{2} \ln 2$	$1 + I_1$	[1]

TRAHS M. EXT 2 TRIAL, 2012

SECTION II SOLUTIONS

MATHEMATICS Extension 2: Question 14.....

Suggested Solutions

Marks

Marker's Comments

(a) Let $S_n = \theta_1 + \theta_2 + \dots + \theta_n < \frac{\pi}{2}$
 Let $P(n): \tan(S_n) \geq \tan \theta_1 + \tan \theta_2 + \dots + \tan \theta_n$

For $P(1)$:

$$\text{LHS} = \tan \theta_1 \quad \text{RHS} = \tan \theta_1$$

$\therefore P(1)$ is true

Assume $P(n)$ is true up to some integer $k \geq 1$

$$\text{i.e. } S_k = \theta_1 + \theta_2 + \dots + \theta_k < \frac{\pi}{2} \quad \text{and}$$

$$\tan(S_k) \geq \tan \theta_1 + \tan \theta_2 + \dots + \tan \theta_k \quad \text{--- (1)}$$

But $P(k+1)$ is true

$$\text{i.e. } \tan(S_{k+1}) \geq \tan \theta_1 + \tan \theta_2 + \dots + \tan \theta_k + \tan \theta_{k+1}$$

Proof $P(k+1)$

$$\begin{aligned} \text{Now } \tan S_{k+1} &= \tan(S_k + \theta_{k+1}) \\ &= \frac{\tan S_k + \tan \theta_{k+1}}{1 - \tan S_k \tan \theta_{k+1}} \end{aligned}$$

$$\geq \frac{\tan \theta_1 + \tan \theta_2 + \dots + \tan \theta_k + \tan \theta_{k+1}}{1 - \tan S_k \tan \theta_{k+1}}$$

* using assumption (1)

$$\text{Now } \theta_1 + \theta_2 + \dots + \theta_k + \theta_{k+1} < \frac{\pi}{2}$$

$$\therefore S_k < \frac{\pi}{2} - \theta_{k+1} \quad 0 \leq \theta_i < \frac{\pi}{2}$$

$$\therefore \tan S_k < \tan\left(\frac{\pi}{2} - \theta_{k+1}\right) = \frac{1}{\tan \theta_{k+1}}$$

$$0 < \tan S_k \tan \theta_{k+1} < 1$$

$$\therefore 0 < 1 - \tan S_k \tan \theta_{k+1} < 1$$

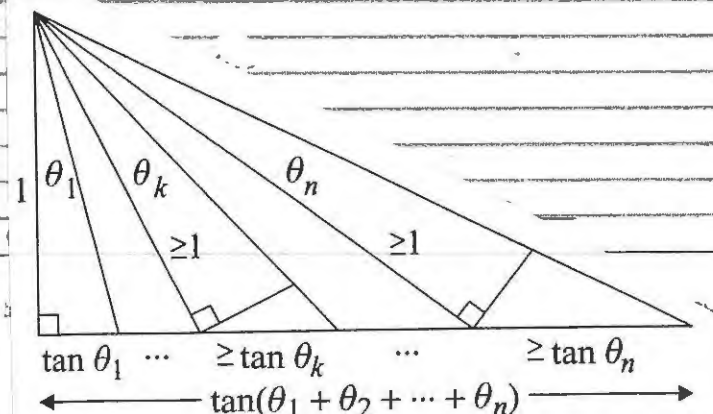
$$\therefore \frac{1}{1 - \tan S_k \tan \theta_{k+1}} > 1$$

$$\text{Hence } \tan S_{k+1} \geq \tan \theta_1 + \tan \theta_2 + \dots + \tan \theta_{k+1}$$

$\therefore P(k+1)$ is true

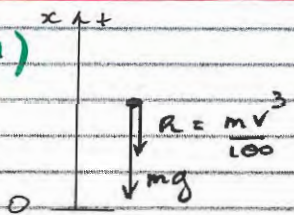
$\therefore$ By the PMI $P(n)$ is true for $n=1, 2, 3, \dots$

NOTE:



TRAHS M. EXT 2 TRIAL, 2012 SECTION II SOLUTIONS

MATHEMATICS Extension 2: Question.....14

Suggested Solutions	Marks	Marker's Comments
(b)(i) $\frac{300x}{100+x^3} = \frac{a}{10+x} + \frac{bx+c}{100-10x+x^2}$ $\therefore 300x \equiv a(100-10x+x^2) + (bx+c)(10+x)$ $x = -10: -3000 = 300a$ $\therefore a = -10$ $x = 0: 0 = 100a + 10c$ $10c = -100a = 1000$ $c = 100$ $x = 10: 3000 = 100a + (10b+c) \times 20$ $= -1000 + 20(10b+100)$ $4000 = 200b + 2000$ $\therefore b = 10$ $\Rightarrow \begin{cases} a = -10 \\ b = 10 \\ c = 100 \end{cases}$		[3]
(ii)(1)  Data $t=0, x=0, v=5, g=10, \ddot{x}=?$ Equation of motion $m\ddot{x} = -mg - \frac{mv^3}{100}$ $\ddot{x} = -10 - \frac{v^3}{100} = -\frac{1}{100}(1000 + v^3)$		[1]
(2) $\ddot{x} = v \frac{dv}{dx} = -\frac{1}{100}(1000 + v^3)$ Max height when $x = H, v = 0$ $\int_5^0 \frac{100v dv}{1000 + v^3} = -\int_0^H dx$ $\frac{10}{3} \int_5^0 \frac{300v dv}{1000 + v^3} = \frac{1}{3} \int_5^0 \frac{-10}{10+v} + \frac{10v+100}{100-10v+v^2} dv \equiv -[x]_0^H$ ie $\frac{10}{3} \int_5^0 \frac{-1}{10+v} + \frac{v+10}{100-10v+v^2} dv = -H$ $\frac{10}{3} \int_5^0 \frac{-1}{10+v} + \frac{\frac{1}{2}(2v-10)}{v^2-10v+100} + \frac{15}{75+(v-5)^2} dv = -H$ $\frac{10}{3} \left[-\ln(10+v) + \frac{1}{2} \ln(v^2-10v+100) + \frac{15}{5\sqrt{3}} \tan^{-1} \frac{(v-5)}{\sqrt{3}} \right]_5^0 = -H$ $-H = \frac{10}{3} \left[\left(-\ln 10 + \frac{1}{2} \ln 100 + \sqrt{3} \tan^{-1} \left(-\frac{1}{\sqrt{3}} \right) \right) - \left(-\ln 15 + \frac{1}{2} \ln 75 + 0 \right) \right]$ $-H = \frac{10}{3} \left[-\sqrt{3} \cdot \frac{\pi}{6} + \ln 15 - \frac{1}{2} \ln 75 \right]$ $H = \frac{10}{3} \left[\frac{\pi\sqrt{3}}{6} + \frac{1}{2} \ln \frac{5}{3} \right] \approx 1.19197...$		[3]
	[8.]	$\therefore$ Max height is <u>1.2 m</u> (1dp)

TRAHS M. EXT 2 TRIAL, 2012 SECTION II SOLUTIONS

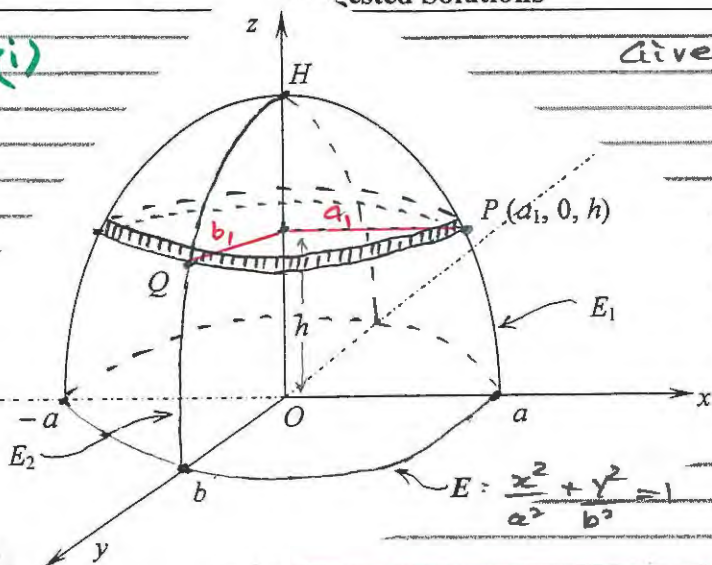
MATHEMATICS Extension 2: Question 14

Suggested Solutions

Marks

Marker's Comments

(i)



Given

$$E_1: \frac{x^2}{a^2} + \frac{z^2}{H^2} = 1$$

so

$$E_2: \frac{y^2}{b^2} + \frac{z^2}{H^2} = 1$$

1

(ii) Since P lies on E_1 at $z=h$

$$\frac{a_1^2}{a^2} + \frac{h^2}{H^2} = 1 \quad \text{ie} \quad a_1^2 = a^2 \left(1 - \frac{h^2}{H^2}\right)$$

And Q lies on E_2 at $z=h$

$$\frac{b_1^2}{b^2} + \frac{h^2}{H^2} = 1 \quad \text{ie} \quad b_1^2 = b^2 \left(1 - \frac{h^2}{H^2}\right)$$

$\therefore$ Area of slice at $z=h$: $A \doteq \pi a_1 b_1$

$$\text{ie } A \doteq \pi a \sqrt{1 - \frac{h^2}{H^2}} \times b \sqrt{1 - \frac{h^2}{H^2}}$$

$$A = \pi ab \left(1 - \frac{h^2}{H^2}\right) \quad \text{qed}$$

2

(iii) Volume of slice $\delta V \doteq \pi ab \left(1 - \frac{h^2}{H^2}\right) \times \delta h$

Let $\delta h \equiv \Delta h$

$$\begin{aligned} \text{Volume of solid } V &\doteq \sum_{h=0}^H \pi ab \left(1 - \frac{h^2}{H^2}\right) \delta h \\ &= \lim_{\delta h \rightarrow 0} \sum_{h=0}^H \pi ab \left(1 - \frac{h^2}{H^2}\right) \delta h \\ &\quad n \rightarrow \infty \quad H \end{aligned}$$

$$\begin{aligned} V &= \pi ab \int_0^H \left(1 - \frac{h^2}{H^2}\right) dh \\ &= \pi ab \left[h - \frac{h^3}{3H^2} \right]_0^H \end{aligned}$$

$$V = \pi ab \left[H - \frac{H^3}{3H^2} \right] = \frac{2}{3} \pi ab H$$

$\therefore$ volume is $\frac{2}{3} ab H \pi$

2

TRAHS M. EXT 2 TRIAL, 2012 SECTION II SOLUTIONS

MATHEMATICS Extension 2: Question.....15

Suggested Solutions	Marks	Marker's Comments
(a) $x^3 + ax^2 + bx - 54 = 0$ (i) As $\alpha^2 + \beta^2 = 0$ — (1) $\alpha^2 + \gamma^2 = 0$ — (2) $\therefore \beta^2 = \gamma^2 \Rightarrow \beta^2 - \gamma^2 = 0$ $\beta = \pm \gamma$ but α, β and γ are distinct $\therefore \beta \neq \gamma$ $\therefore \beta = -\gamma \Rightarrow \beta + \gamma = 0$		[2]
(ii) Now $\Delta_1 = \sum \alpha = \alpha + \beta + \gamma = -a$ ie $\alpha + 0 = -a$ $\alpha = -a$ As a is real $\therefore \alpha$ is real <i>(date)</i>		[2]
(iii) Now $\alpha^2 + \beta^2 = 0$ $\beta^2 = -\alpha^2 = i^2 \alpha^2$ $\beta = \pm i\alpha$ As $\alpha \in \mathbb{R}$ $\therefore \beta$ is purely imaginary As $\gamma = -\beta = \mp i\alpha$. So again γ is purely imaginary		[2]
(iv) Now $\Delta_3 = \sum \alpha\beta\gamma = \alpha\beta\gamma = -(-54) = 54$ but $\gamma = -\beta \therefore -\alpha\beta^2 = 54$ but $\beta^2 = -\alpha^2 \therefore \alpha^3 = 54$ $\alpha = \sqrt[3]{54} \in \mathbb{R}$ $\therefore \alpha = -\alpha = -\sqrt[3]{54} = -\sqrt[3]{2^3 \cdot 3^3} = -(54)^{\frac{1}{3}}$ $\Delta_2 = \sum \alpha\beta = \alpha\beta + \beta\gamma + \gamma\alpha = b$ $\alpha(\beta + \gamma) + \beta\gamma = b$ $\Rightarrow \beta(-\beta) = b$ $-\beta^2 = b$ $+\alpha^2 = b$ $\therefore b = (-\sqrt[3]{54}) = \sqrt[3]{54^2} = 54^{\frac{2}{3}}$ So $\alpha = -\sqrt[3]{54}$ and $b = \sqrt[3]{54^2}$		[2]

TRAHS M. EXT 2 TRIAL, 2012 SECTION II SOLUTIONS

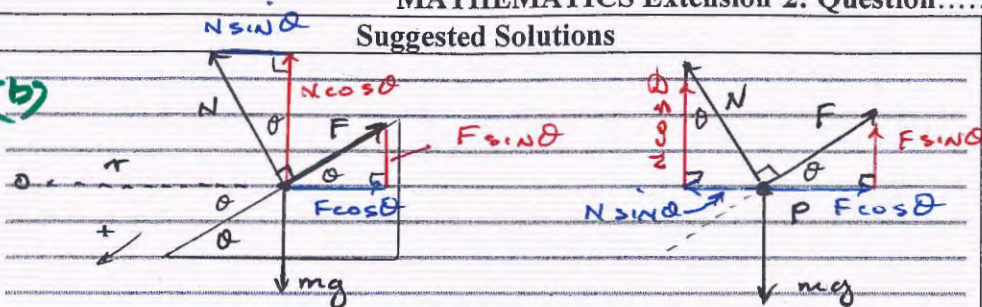
MATHEMATICS Extension 2: Question 15

Suggested Solutions

Marks

Marker's Comments

(b)



(i) Given vertical resolution at P is

$$N \cos \theta + F \sin \theta = mg$$

HORIZONTAL is $N \sin \theta - F \cos \theta = \frac{mv^2}{r}$

(Newton's II)

1

(ii) $r = 80$, $\theta = 45^\circ$, $g = 10$ $F \leq \frac{N}{9}$
So min. speed when $F = \frac{N}{9}$ or $N = 9F$

$$\therefore \frac{9F}{\sqrt{2}} + \frac{F}{\sqrt{2}} = m \times 10$$

$$F = m\sqrt{2} \text{ and}$$

$$\frac{9F}{\sqrt{2}} - \frac{F}{\sqrt{2}} = \frac{mv^2}{80}$$

$$\therefore mv^2 = 800F = 640m$$

$$\therefore v^2 = 640$$

$$v = 8\sqrt{10}$$

min speed is $8\sqrt{10} \text{ m/s}$

[91.1 kmph]

$$\frac{8F}{\sqrt{2}} = \frac{mv^2}{80}$$

$$mv^2 = \frac{640F}{\sqrt{2}}$$

2

(iii) For upwards motion

VERTICAL RESOLUTION: $N \cos \theta - F \sin \theta = mg$

$$\text{ie } \frac{N}{\sqrt{2}} - \frac{F}{\sqrt{2}} = 10m$$

$$\text{ie } \frac{8F}{\sqrt{2}} = 10m$$

$$\text{ie } F = \frac{10m\sqrt{2}}{8}$$

HORIZONTAL: $\frac{N}{\sqrt{2}} + \frac{F}{\sqrt{2}} = \frac{mv^2}{80}$

$$mv^2 = 800F = \frac{800 \times 10 \times m\sqrt{2}}{8} = 1000m$$

$$v^2 = 1000$$

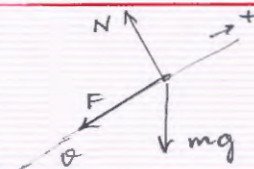
$$v = \sqrt{1000} = 10\sqrt{10}$$

max speed is $10\sqrt{10} \text{ m/s}$

[113.8 kmph]

$$\frac{10F}{\sqrt{2}} = \frac{mv^2}{80}$$

3



(iv)

when no slipping $F = 0$ so from (ii) or (iii)

$$\frac{N}{\sqrt{2}} = 10m \text{ and } \frac{N}{\sqrt{2}} = \frac{mv^2}{80}$$

$$\Rightarrow v^2 = 800 \text{ ie } v = 20\sqrt{2}$$

[101.8 kmph]

• can show $v^2 = v_{\min} \times v_{\max}$
and/or use $\tan \theta = \frac{v^2}{rg}$

4

(11.)

$\therefore$ the optimum speed for no slippage at 45° is $20\sqrt{2} \text{ m/s}$

MATHEMATICS Extension 2: Question 16

Suggested Solutions

Marks

Marker's Comments

(a) $I = \int_0^1 \frac{\ln(1+x)}{1+x^2} dx$

$$x = \frac{1-u}{1+u}$$

x	u
1	0
0	1

$$\frac{dx}{du} = \frac{-1(1+u) - (1-u) \times 1}{(1+u)^2}$$

$$= \frac{-1-u-1+u}{(1+u)^2} = \frac{-2}{(1+u)^2}$$

$$dx = \frac{-2}{(1+u)^2} du$$

$$1+x = 1 + \frac{1-u}{1+u} = \frac{2}{1+u}$$

$$1+x^2 = 1 + \frac{(1-u)^2}{(1+u)^2} = \frac{1+2u+u^2+1-2u+u^2}{(1+u)^2}$$

$$1+x^2 = \frac{2(1+u^2)}{(1+u)^2}$$

$$\therefore I = \int_1^0 \frac{\ln \frac{2}{1+u} \times \frac{-2}{(1+u)^2} du}{\frac{2(1+u^2)}{(1+u)^2}}$$

$$= \int_1^0 \frac{-2 \ln \frac{2}{1+u}}{2(1+u^2)} du$$

$$I = + \int_0^1 \frac{\ln 2}{1+u^2} - \frac{\ln(1+u)}{1+u^2} du$$

$$= \int_0^1 \frac{\ln 2}{1+x^2} dx - I$$

u is a dummy variable

$$2I = \ln 2 \cdot [\tan^{-1} x]_0^1 = \ln 2 \cdot \left(\frac{\pi}{4}\right)$$

$$\therefore I = \ln 2 \cdot \frac{\pi}{8} \text{ of eval.}$$

4

(b) $x^4 + 3x - 1 = 0$

METHOD I

$$\text{so } \alpha^4 + 3\alpha - 1 = 0$$

$$\text{so } \sum_{i=1}^4 \alpha_i^4 + 3 \sum_{i=1}^4 \alpha_i - \sum_{i=1}^4 1 = 0$$

$$\sum \alpha^4 = -3(\alpha + \beta + \gamma + \delta) + 4 \times 1 = -3 \times 0 + 4$$

$$\alpha^4 + \beta^4 + \gamma^4 + \delta^4 = 4$$

Method II
construct Polyn
whose roots are
 $\alpha^4, \beta^4, \gamma^4, \delta^4$
ie $81x = (1-x)^4$

$$\text{ie } x^4 - 4x^3 + 6x^2 - 8x + 1 = 0$$

Hence ...

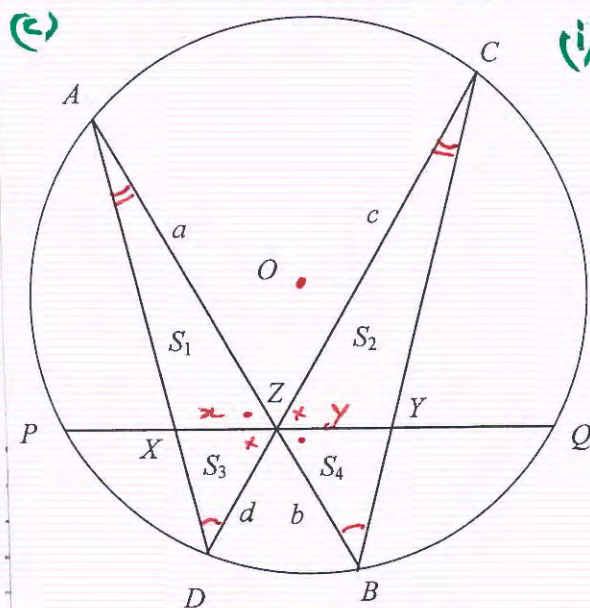
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MATHEMATICS Extension 2: Question 16

Suggested Solutions

Marks

Marker's Comments



$$S_1 = \frac{1}{2} a \cdot AX \sin A$$

$$S_2 = \frac{1}{2} c \cdot CY \sin C$$

As $\angle C = \angle A$ (angles subtended at circumference are equal standing on arc DB)

$$\therefore S_1 = \frac{a \cdot AX}{c \cdot CY} \quad \text{QED}$$

$$S_3 = \frac{1}{2} d \cdot x \sin \hat{XZA}$$

$$S_4 = \frac{1}{2} b \cdot y \sin \hat{BZY}$$

but $\angle BZY = \angle XZA$ (vertically opposite angles are equal)

$$\therefore \frac{S_1}{S_4} = \frac{a \cdot x}{b \cdot y} \quad \text{QED}$$

3

(ii) Similarly $\angle D = \angle B$ (angles subtended at circumference are equal standing on arc AC)

$$\text{so } \frac{S_3}{S_4} = \frac{\frac{1}{2} DX \cdot d \sin D}{\frac{1}{2} BY \cdot b \sin B} = \frac{d \cdot XD}{b \cdot YB}$$

$$\text{so } \frac{S_1}{S_2} \times \frac{S_3}{S_4} = \frac{a \cdot AX}{c \cdot CY} \times \frac{d \cdot XD}{b \cdot YB} = \frac{a \cdot d \cdot AX \cdot XD}{b \cdot c \cdot CY \cdot YB}$$

$$\text{Now } \frac{S_3}{S_2} = \frac{\frac{1}{2} \cdot d \cdot x \sin \hat{DXZ}}{\frac{1}{2} \cdot c \cdot y \sin \hat{CYZ}} = \frac{d \cdot x}{c \cdot y}$$

$$\therefore \frac{S_1}{S_4} \times \frac{S_3}{S_2} = \frac{a \cdot x}{b \cdot y} \times \frac{d \cdot x}{c \cdot y} = \frac{a \cdot d \cdot x^2}{b \cdot c \cdot y^2} \quad \text{QED}$$

2

$$(iii) \therefore \frac{a \cdot d \cdot AX \cdot XD}{b \cdot c \cdot CY \cdot YB} = \frac{a \cdot d \cdot x^2}{b \cdot c \cdot y^2}$$

$$\text{so } \frac{x^2}{y^2} = \frac{AX \cdot XD}{CY \cdot YB} = \frac{PX \cdot XQ}{PY \cdot YQ} = \frac{(p-x)(x+q)}{(p+y)(q-y)}$$

(The products of the intercepts of 2 intersecting chords are equal)

1

MATHEMATICS Extension 2: Question.....16

Suggested Solutions	Marks	Marker's Comments
(c) (iv) $\frac{x^2}{y^2} = \frac{(p-x)(x+q)}{(p+y)(q-y)}$ $= \frac{px + pq - x^2 - qx}{pq - py + qy - y^2}$ $x^2pq - x^2yp + x^2yq - x^2y^2 = xy^2p + y^2pq - x^2y^2 - xy^2q$ $pq(x^2 - y^2) = xyp - x^2yq + xy^2p - xy^2q$ $pq(x-y)(x+y) = x^2y(p-q) + xy^2(p-q)$ $pq(x-y)(x+y) = xy(p-q)(x+y)$ $\therefore pq(x-y) = xy(p-q)$ $\text{ie } \frac{x-y}{xy} = \frac{p-q}{pq}$ $\therefore \frac{1}{y} - \frac{1}{x} = \frac{1}{q} - \frac{1}{p} \Rightarrow \frac{1}{x} - \frac{1}{y} = \frac{1}{p} - \frac{1}{q}$		2
(v) If Z is the mid point $\therefore p = q$ $\therefore \frac{1}{p} - \frac{1}{q} = 0$ $\Rightarrow \underline{x = y}$		11